Degeneracy Alone Does Not Force a Square-Root LP-Relaxation Gap in Finite-Horizon Restless Bandits

June 23, 2026

Abstract

Gast, Gaujal, and Yan conjecture that, for "essentially all" degenerate finite-horizon restless bandits, LP-compatible policies have an LP-relaxation gap of order $N^{-1/2}$. This note gives a small explicit degenerate restless-bandit instance for which an LP-compatible priority policy has gap 0 for every even N and gap $1/(4N)$ for every odd N, under the usual randomized-budget convention. Thus degeneracy by itself does not imply an $\Omega(N^{-1/2})$ lower bound. The result is therefore a solution of the literal degeneracy-alone formulation of the lower-bound question, but only partial progress toward the authors' qualified generic conjecture, because the phrase "essentially all" requires an additional genericity or measure-theoretic hypothesis that this example is not intended to address.

1 Introduction and status of the result

The linear-programming relaxation for finite-horizon restless bandits gives an upper bound on the optimal N-arm value. Gast, Gaujal, and Yan [1] analyze when LP-compatible policies approach this relaxation value at square-root or exponential rates. In their discussion of degenerate instances, they conjecture that a square-root gap is typical for LP-compatible policies. The open-problems formulation [2] asks, in particular, whether one can prove or disprove an $\Omega(N^{-1/2})$ lower bound for LP-compatible policies on degenerate finite-horizon restless bandits.

The purpose of this note is to separate two possible interpretations of that question. If the claim is interpreted literally as

$$\text{degenerate} \quad \Longrightarrow \quad \text{every LP-compatible policy has gap } \Omega(N^{-1/2}),$$

then the claim is false. We exhibit a three-state, two-period instance that is degenerate in the stated LP sense, but admits an LP-compatible Markov priority policy whose finite-N gap is $O(1/N)$, and is in fact zero on the infinite subsequence of even system sizes.

This does not settle the original generic formulation. The authors' wording is qualified by "essentially all", and no measure or genericity class is specified in the open-problems extraction. The example below should instead be read as a sharp obstruction to any theorem whose hypotheses are only "finite horizon" and "degenerate". Any correct square-root lower-bound theorem must also exclude examples in which the degenerate boundary is reached with no central-limit-scale fluctuation in the relevant direction.

2 The LP relaxation and terminology

We recall the finite-horizon model only to fix notation. There is a finite state space $\mathcal{S}$, two actions $a \in \{0,1\}$, transition matrices $P^a = (P^a_{s,s'})_{s,s' \in \mathcal{S}}$, rewards R^a_s, activation fraction $\alpha \in (0,1)$, horizon

T, and initial distribution $m(0) \in \Delta(\mathcal{S})$. For the LP relaxation, the decision variables are state-action masses $y_{s,a}(t)$, where $t = 0, \dots, T-1$. The relaxation is

$$V_{\mathrm{rel}}(m(0), T) = \max_{y \geq 0} \quad \sum_{t=0}^{T-1} \sum_{s \in \mathcal{S}} \sum_{a \in \{0,1\}} R_s^a y_{s,a}(t) \tag{1}$$

$$\text{subject to} \quad y_{s,0}(0) + y_{s,1}(0) = m_s(0), \qquad s \in \mathcal{S}, \tag{2}$$

$$y_{s,0}(t+1) + y_{s,1}(t+1) = \sum_{r \in \mathcal{S}} \sum_{a \in \{0,1\}} y_{r,a}(t) P_{r,s}^a, \quad s \in \mathcal{S},\ t = 0, \dots, T-2, \tag{3}$$

$$\sum_{s \in \mathcal{S}} y_{s,1}(t) = \alpha, \qquad t = 0, \dots, T-1. \tag{4}$$

Let

$$m_s^*(t) := y_{s,0}^*(t) + y_{s,1}^*(t)$$

for an optimal LP solution y^*. A Markov policy $\pi = (\pi_t)_{t=0}^{T-1}$ at the fluid level is an admissible map

$$\pi_t : \Delta(\mathcal{S}) \to \mathbb{R}_+^{\mathcal{S} \times \{0,1\}}$$

such that, for every $m \in \Delta(\mathcal{S})$,

$$\sum_{a \in \{0,1\}} \pi_t(m)_{s,a} = m_s, \qquad \sum_{s \in \mathcal{S}} \pi_t(m)_{s,1} = \alpha.$$

It is LP-compatible if there is an optimal LP solution y^* such that

$$\pi_t(m^*(t)) = y^*(t), \qquad t = 0, \dots, T-1.$$

For an optimal solution y^* define the set of split states at time t by

$$S_0(t; y^*) = \{s \in \mathcal{S} : y_{s,0}^*(t) > 0 \text{ and } y_{s,1}^*(t) > 0\}.$$

The instance is degenerate, in the sense relevant here, if for every optimal LP solution there is at least one time t such that $S_0(t; y^*) = \varnothing$.

3 The counterexample

Set

$$\mathcal{S} = \{x, g, b\}, \qquad \alpha = \frac{1}{2}, \qquad T = 2, \qquad m(0) = \delta_x.$$

The transitions are deterministic. From x, action 1 sends the arm to g, while action 0 sends the arm to b:

$$P_{x,g}^1 = 1, \qquad P_{x,b}^0 = 1.$$

The states g and b are absorbing under both actions:

$$P_{g,g}^a = 1, \qquad P_{b,b}^a = 1, \qquad a \in \{0,1\}.$$

All other transition probabilities are zero. The only positive reward is

$$R_g^1 = 1,$$

and

$$R_s^a = 0 \qquad \text{for every other pair } (s,a).$$

Thus the reward is not constant, and the transition out of x depends on the action.

4 LP value and degeneracy

Lemma 1. *The LP relaxation for the instance above has a unique optimal solution. Its value is*

$$V_{\mathrm{rel}}(m(0), 2) = \frac{1}{2}.$$

Moreover the instance is degenerate: at time 1 *no state is split between the two actions in the unique optimal LP solution.*

Proof. At time 0, the initial condition gives

$$y_{x,0}(0) + y_{x,1}(0) = 1, \qquad y_{g,0}(0) = y_{g,1}(0) = y_{b,0}(0) = y_{b,1}(0) = 0.$$

The budget constraint at time 0 then forces

$$y^*_{x,1}(0) = \frac{1}{2}, \qquad y^*_{x,0}(0) = \frac{1}{2}.$$

There is no reward at time 0, since $R^0_x = R^1_x = 0$.

By the deterministic transition rule from x, the time-1 fluid state is

$$m^*_g(1) = \frac{1}{2}, \qquad m^*_b(1) = \frac{1}{2}, \qquad m^*_x(1) = 0.$$

At time 1, the LP objective is simply $y_{g,1}(1)$, because $R^1_g = 1$ and all other rewards are zero. The state-mass and budget constraints imply

$$y_{g,1}(1) \le m^*_g(1) = \frac{1}{2}, \qquad y_{g,1}(1) + y_{b,1}(1) = \frac{1}{2}.$$

Therefore the maximum possible value is 1/2, and it is achieved only if

$$y^*_{g,1}(1) = \frac{1}{2}, \qquad y^*_{b,1}(1) = 0.$$

The remaining state-mass constraints then force

$$y^*_{g,0}(1) = 0, \qquad y^*_{b,0}(1) = \frac{1}{2}, \qquad y^*_{x,0}(1) = y^*_{x,1}(1) = 0.$$

Thus the optimal LP solution is unique and has value 1/2. At time 1, neither g nor b nor x has positive mass under both actions. Hence $S_0(1; y^*) = \varnothing$, proving degeneracy. □

5 An LP-compatible priority policy

We now define a priority policy carefully enough to remove any tie-breaking ambiguity. For an ordering $O = (o_1, o_2, o_3)$ of $\{x, g, b\}$ and $m \in \Delta(\mathcal{S})$, define active masses recursively by

$$A^O_{o_k}(m) := \min\left\{ m_{o_k}, \left[\alpha - \sum_{\ell<k} m_{o_\ell}\right]_+ \right\}, \qquad k = 1, 2, 3.$$

Equivalently, $A^O(m)$ fills the activation budget greedily according to the priority order O. Since $\sum_s m_s = 1$ and $\alpha = 1/2$, one has $\sum_s A^O_s(m) = \alpha$. Define

$$\Pi^O(m)_{s,1} = A^O_s(m), \qquad \Pi^O(m)_{s,0} = m_s - A^O_s(m).$$

This is an admissible piecewise-affine Markov decision rule.

Let

$$O_0 = (x, b, g), \qquad O_1 = (g, b, x),$$

and define

$$\pi_0 = \Pi^{O_0}, \qquad \pi_1 = \Pi^{O_1}.$$

At the LP trajectory, $m^*(0) = \delta_x$, and hence

$$\pi_0(m^*(0))_{x,1} = \frac{1}{2}, \qquad \pi_0(m^*(0))_{x,0} = \frac{1}{2},$$

with all other time-0 state-action masses equal to zero. Also $m_g^*(1) = m_b^*(1) = 1/2$, and the priority order $O_1 = (g, b, x)$ gives

$$\pi_1(m^*(1))_{g,1} = \frac{1}{2}, \qquad \pi_1(m^*(1))_{g,0} = 0, \qquad \pi_1(m^*(1))_{b,1} = 0, \qquad \pi_1(m^*(1))_{b,0} = \frac{1}{2}.$$

Thus π reproduces the unique optimal LP solution along the LP trajectory.

Lemma 2. *The priority policy π is LP-compatible.*

Proof. The displayed identities show that $\pi_t(m^*(t)) = y^*(t)$ for $t = 0, 1$, where y^* is the optimal LP solution from Lemma 1. □

6 Finite-N implementation and exact gap

We use the randomized-budget convention stated in the problem formulation: at each time t, if αN is not an integer, the policy randomizes between $\lfloor \alpha N \rfloor$ and $\lfloor \alpha N \rfloor + 1$ active arms so that the expected number of active arms is αN. For $\alpha = 1/2$, this means the time-t budget is

$$B_t^{(N)} = \begin{cases} N/2, & N \text{ even}, \\ \lfloor N/2 \rfloor, & N \text{ odd, with probability } 1/2, \\ \lceil N/2 \rceil, & N \text{ odd, with probability } 1/2. \end{cases}$$

For odd N, take the randomizations at the two times to be independent. Given the realized budget $B_t^{(N)}$ and the empirical state counts, the finite-N implementation activates arms greedily according to the same priority orders O_0 and O_1 as above.

Let K_N be the number of arms activated at time 0. Initially all arms are in state x, so

$$K_N = B_0^{(N)}.$$

The K_N activated arms move deterministically to g, and the remaining $N - K_N$ arms move deterministically to b. Therefore the number of arms in state g at time 1 is exactly K_N.

At time 1, the priority order is g first. Thus the number of rewarding active arms is

$$\min\{K_N, B_1^{(N)}\}.$$

There is no reward at time 0, so the per-arm expected value is

$$V_\pi^{(N)}(m(0), 2) = \frac{1}{N} \mathbb{E}\big[\min\{K_N, B_1^{(N)}\}\big]. \tag{5}$$

If $N = 2n$ is even, then $K_N = B_1^{(N)} = n$, so (5) gives

$$V_\pi^{(N)}(m(0), 2) = \frac{n}{2n} = \frac{1}{2}.$$

If $N = 2n + 1$ is odd, then $K_N = n$ with probability $1/2$ and $K_N = n + 1$ with probability $1/2$. Conditional on $K_N = n$, all n good arms are activated at time 1, regardless of $B_1^{(N)}$. Conditional on $K_N = n + 1$, the number of activated good arms is $B_1^{(N)}$, whose expectation is $n + 1/2$. Hence

$$\begin{aligned} V_\pi^{(N)}(m(0), 2) &= \frac{1}{2n+1}\left(\frac{1}{2}n + \frac{1}{2}\left(n + \frac{1}{2}\right)\right) \\ &= \frac{1}{2} - \frac{1}{4(2n+1)} = \frac{1}{2} - \frac{1}{4N}. \end{aligned}$$

Combining this with Lemma 1, we obtain the exact gap formula

$$V_{\mathrm{rel}}(m(0), 2) - V_\pi^{(N)}(m(0), 2) = \begin{cases} 0, & N \text{ even}, \\ \dfrac{1}{4N}, & N \text{ odd}. \end{cases} \tag{6}$$

7 Main consequence

Theorem 1. *There exists a degenerate finite-horizon restless-bandit instance with nonconstant rewards and action-dependent transitions, and an LP-compatible admissible Markov priority policy π, such that*

$$0 \le V_{\mathrm{rel}}(m(0), 2) - V_\pi^{(N)}(m(0), 2) \le \frac{1}{4N} \qquad \textit{for every } N.$$

Moreover the gap is exactly zero for every even N. Consequently, there is no constant $\underline{C} > 0$ such that

$$\frac{\underline{C}}{\sqrt{N}} \le \left| V_\pi^{(N)}(m(0), 2) - V_{\mathrm{rel}}(m(0), 2) \right|$$

for all sufficiently large N for this instance and this LP-compatible policy.

Proof. The instance is degenerate by Lemma 1. The policy is LP-compatible by Lemma 2. The exact finite-N gap is given by (6). In particular, the gap is zero on the infinite subsequence $N = 2, 4, 6, \ldots$. Therefore no positive multiple of $N^{-1/2}$ can lower-bound the gap for all sufficiently large N. Even on the complementary odd subsequence, the gap is $1/(4N) = o(N^{-1/2})$. □

Remark 1 (Integral budgets)**.** *If one avoids randomized budgets and considers only system sizes for which αN is an integer, then here N is even and the gap is identically zero for every admissible system size. Thus the example does not rely on the randomized-budget convention.*

Remark 2 (Why the example is exceptional)**.** *The degenerate boundary at time 1 is real: the optimal LP solution activates all of the g mass and none of the b mass. However, the amount of g mass at time 1 is not subject to binomial or central-limit-scale noise. It is exactly the number of arms deliberately activated at time 0, and the transition from x to g under action 1 is deterministic. Thus the only finite-N discrepancy comes from the deterministic or randomized rounding of the budget, which is of order $1/N$ in per-arm units. This identifies a missing hypothesis in any possible square-root lower-bound theorem: degeneracy must be accompanied by a nonzero fluctuation in the boundary-crossing direction.*

8 Conclusion

The construction above proves that degeneracy alone is not sufficient to force an $\Omega(N^{-1/2})$ LP-relaxation gap for LP-compatible policies. In that precise literal sense, it resolves the lower-bound question negatively.

It should not be presented as a complete disproof of the authors' qualified conjecture for "essentially all" degenerate restless bandits. Rather, it is partial progress toward the intended generic question: it shows that any true generic theorem must add hypotheses excluding deterministic, zero-transverse-variance degeneracies of the kind used here.

References

[1] Nicolas Gast, Bruno Gaujal, and Chen Yan. Linear Program-Based Policies for Restless Bandits: Necessary and Sufficient Conditions for (Exponentially Fast) Asymptotic Optimality. *Mathematics of Operations Research* 49(4):2468–2491, 2024. doi:10.1287/moor.2022.0101; arXiv:2106.10067.

[2] Pranav Nuti, Eric Fithian, and Rad Niazadeh, maintainers. Open Problems in Operations Research, Problem W4389224581_p0: "Prove square-root lower bound for LP-compatible policies in degenerate restless bandits." Generated/unverified problem extraction, accessed June 23, 2026. https://pranav-nuti.github.io/open-problems-in-or/problem.html?id=W4389224581_p0&page=9&n=101.